

Safety of sildenafil citrate: review of 67 double-blind placebo-controlled trials and the postmarketing safety database



Aim: To review special safety topics associated with sildenafil and to document the tolerability of 50- and 100-mg doses, overall and by age, in men with erectile dysfunction (ED). Methods: Data were collated from 67 double-blind placebo-controlled (DBPC) trials (> 14,000 men) conducted by the manufacturer and from the manufacturer's postmarketing safety database (39,277 patients). The DBPC data were stratified by dose, starting dose and age (≥ 65 and ≥ 75 years). Special safety topics included cardiovascular risk, priapism, non-arteritic anterior ischaemic optic neuropathy (NAION), impaired renal and hepatic function, drug interactions (i.e. nitrates, cytochrome P3A4 inhibitors, other ED therapies and α -blockers) and incorrect use. Results: Sildenafil was well tolerated at a dose of 50 or 100 mg in men with ED, overall, in those aged ≥ 65 years, and in those aged ≥ 75 years. Analyses of the databases did not reveal any causal link between sildenafil and cardiovascular events, or any new safety risks relating to cardiovascular events, priapism, NAION, hearing loss or drug interactions. In the small number of men with moderate impairment of renal function or hepatic function who were treated with sildenafil in DBPC trials, the safety profile was similar to that in men with no impairment of renal or hepatic function. Overdose with sildenafil was rare in the ED population. No new safety issues, emerging trends or adverse reactions were identified in conjunction with overdose, dependence, abuse or misuse. Conclusion: This collated review confirms generally the good tolerability and established safety profile of sildenafil 50 and 100 mg in men with ED and reveals no new safety issues.